

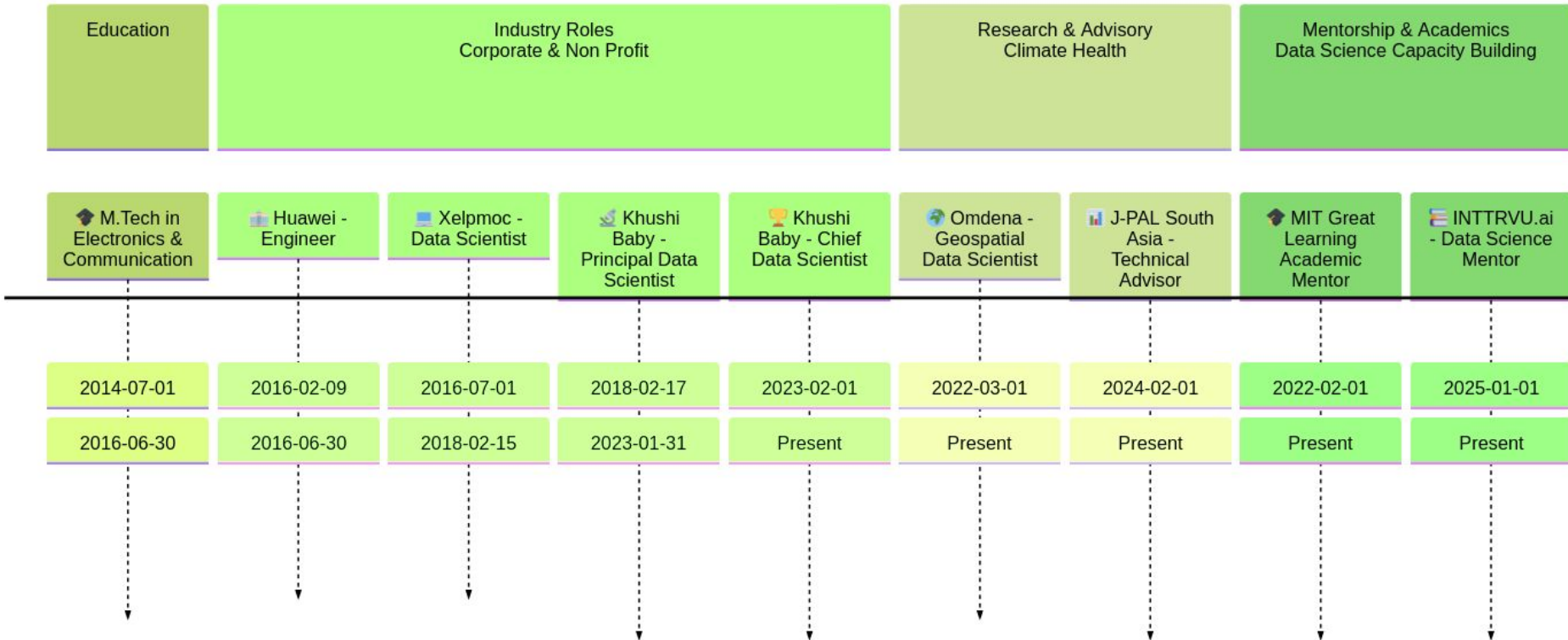
Public Health Informatics: Data, Decisions, and Career Pathway

Mohammad Sarfarazul Ambiya
Chief Data Scientist

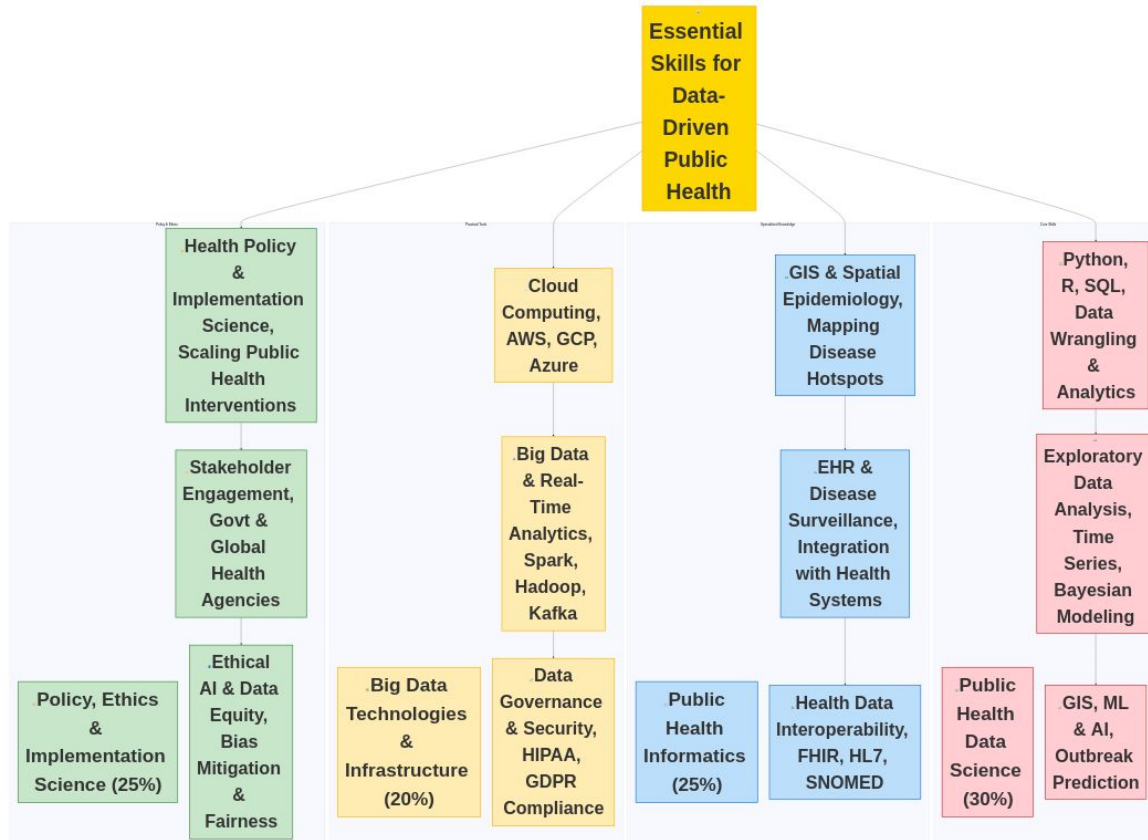
Session Outline

- About me
- Essential skills to excel in public health informatics
- Data Science Process
- Why Spatial & Time Series?
- Khushi Baby Public Health Informatics

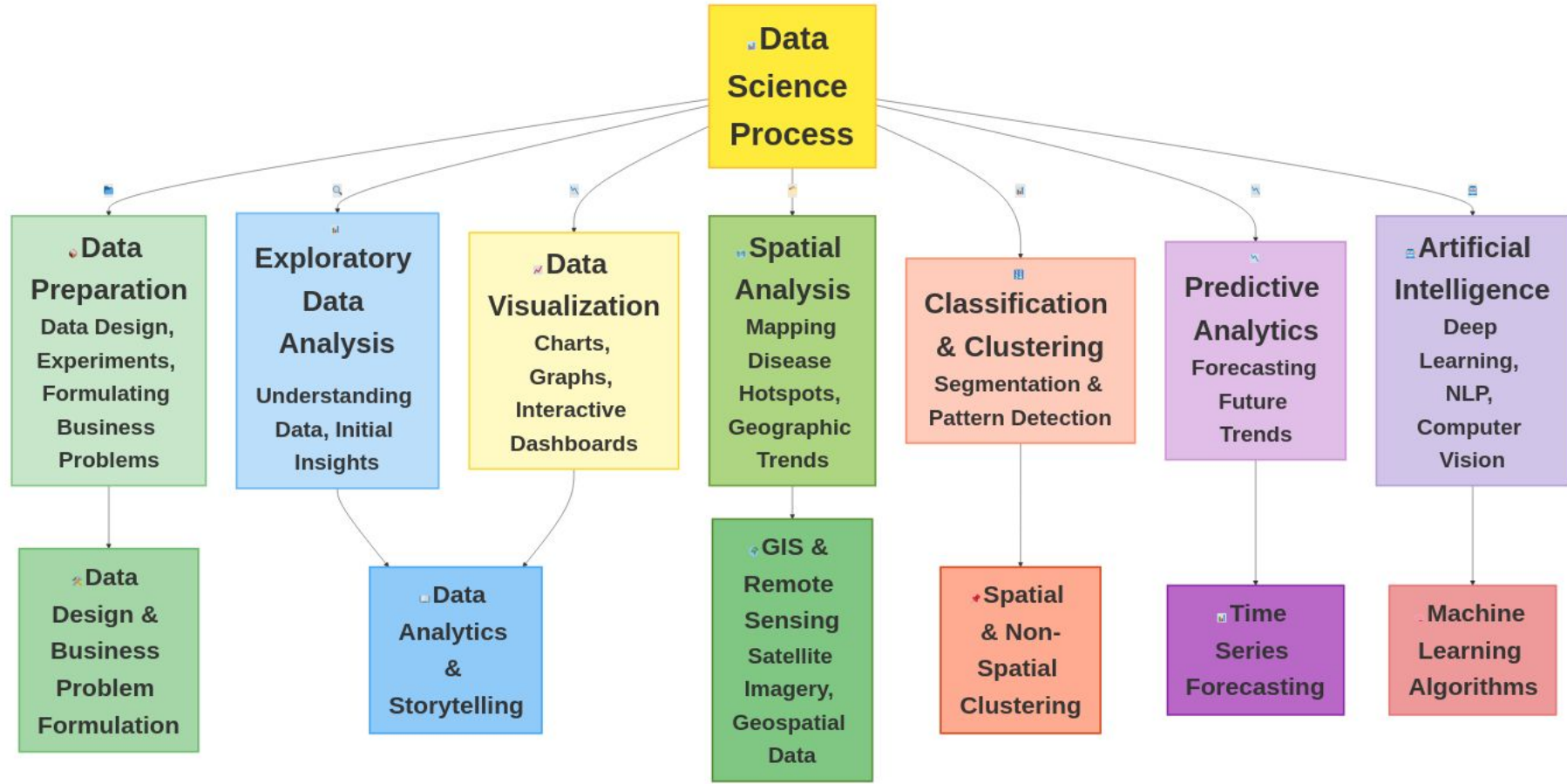
About me



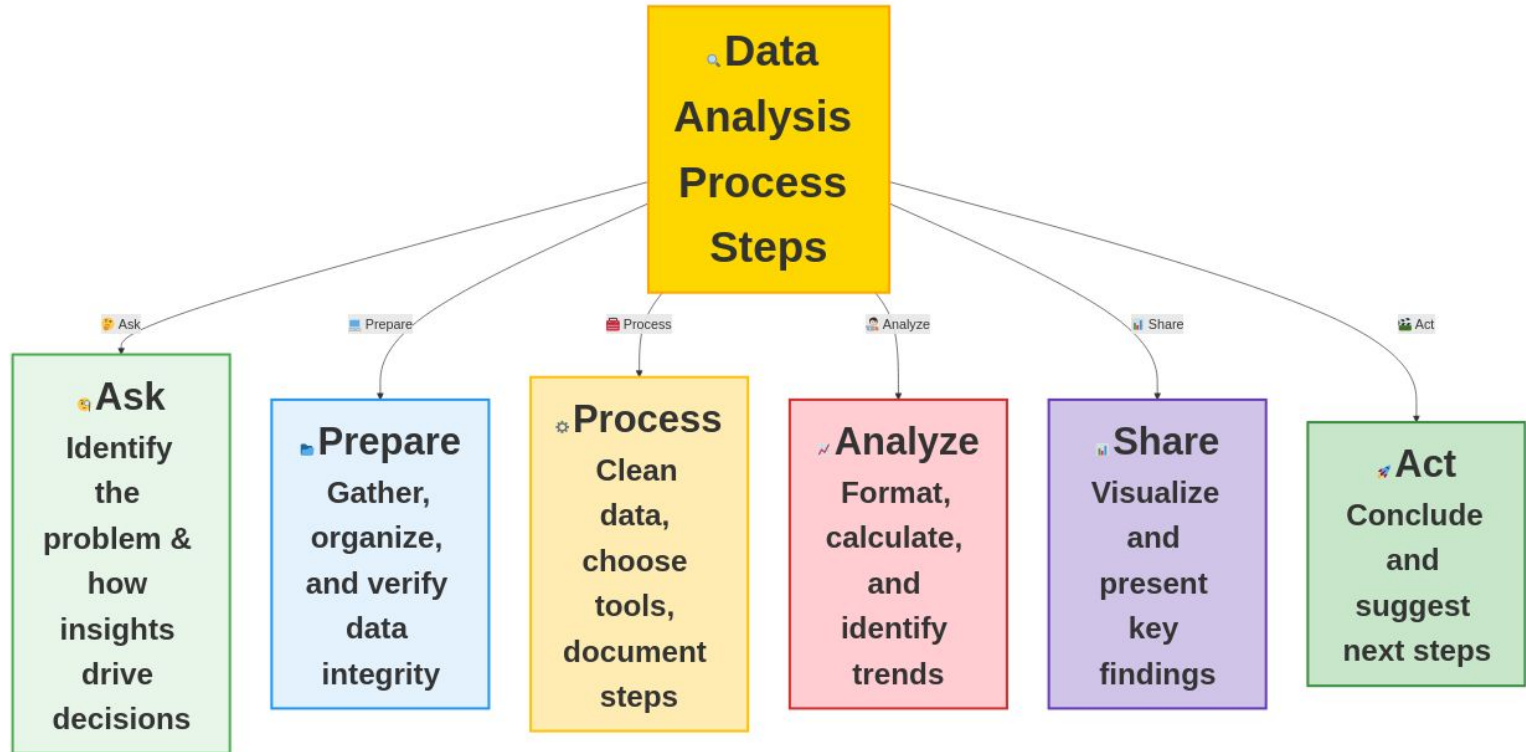
Essential Skills for Excelling in Public Health - at least 70%



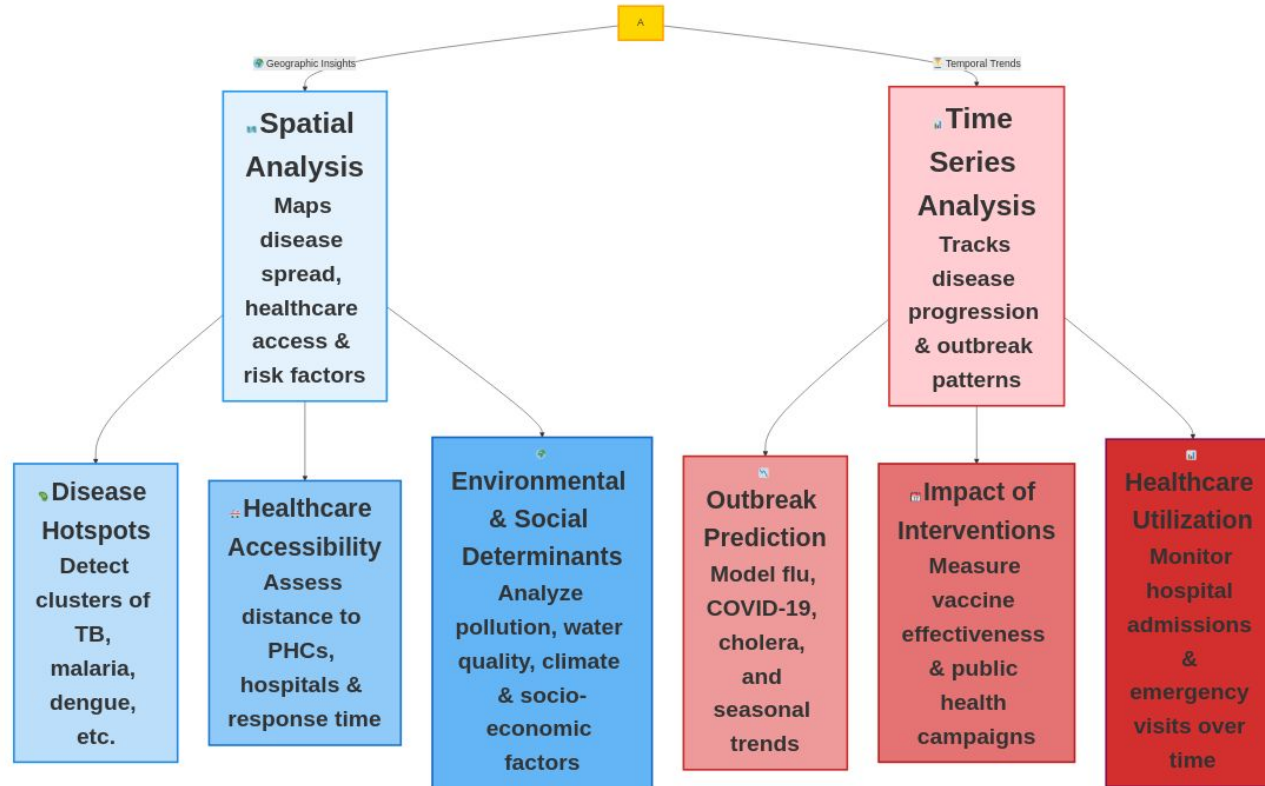
General Data Science Process



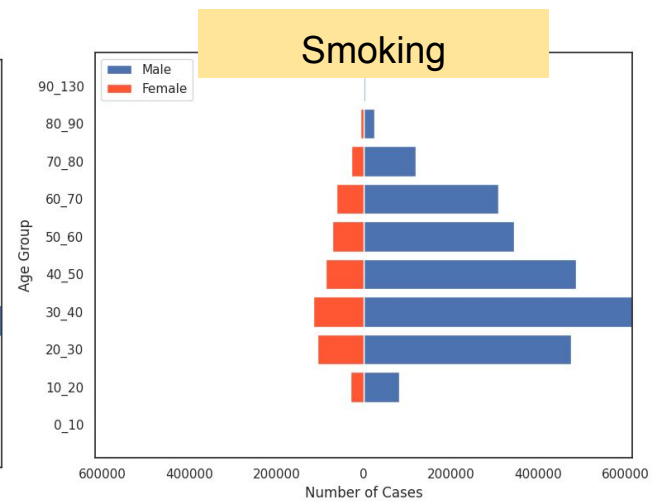
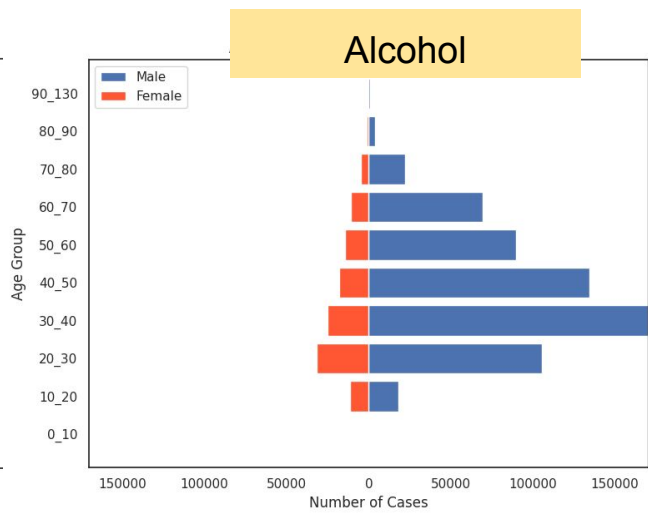
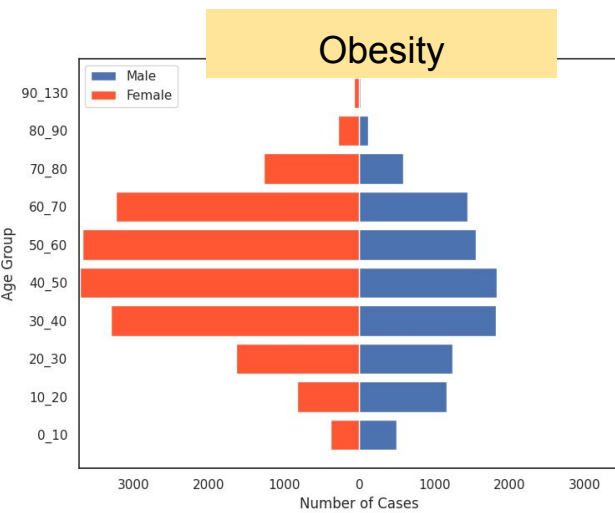
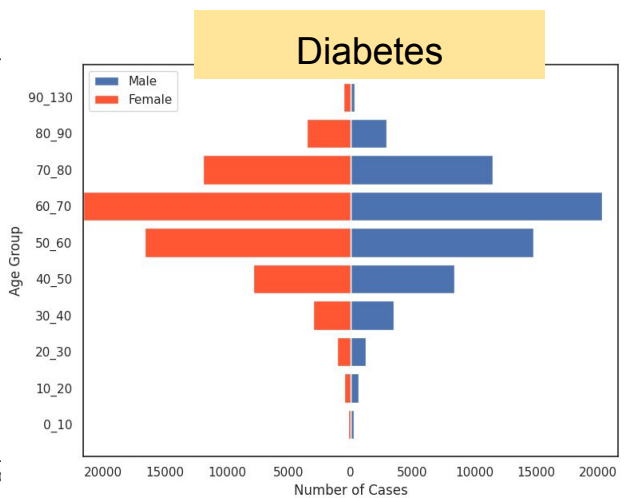
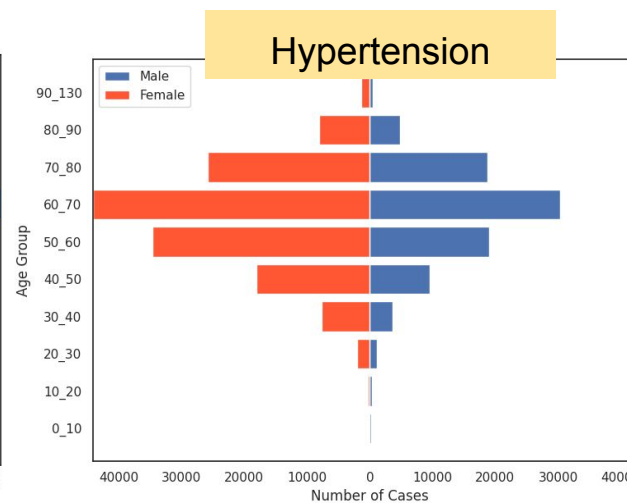
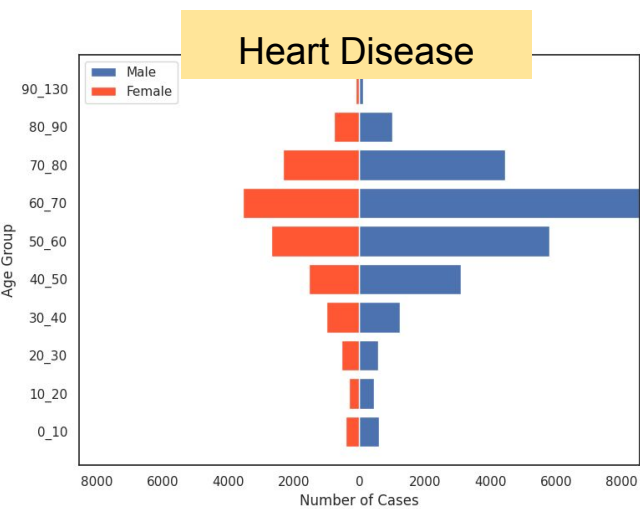
Steps in Data Analysis



Why spatial & time series analysis the most important in public health



Khushi Baby Public Health Informatics



70K

CHWs

48M

Beneficiaries
reached

6M

High-risk
identified

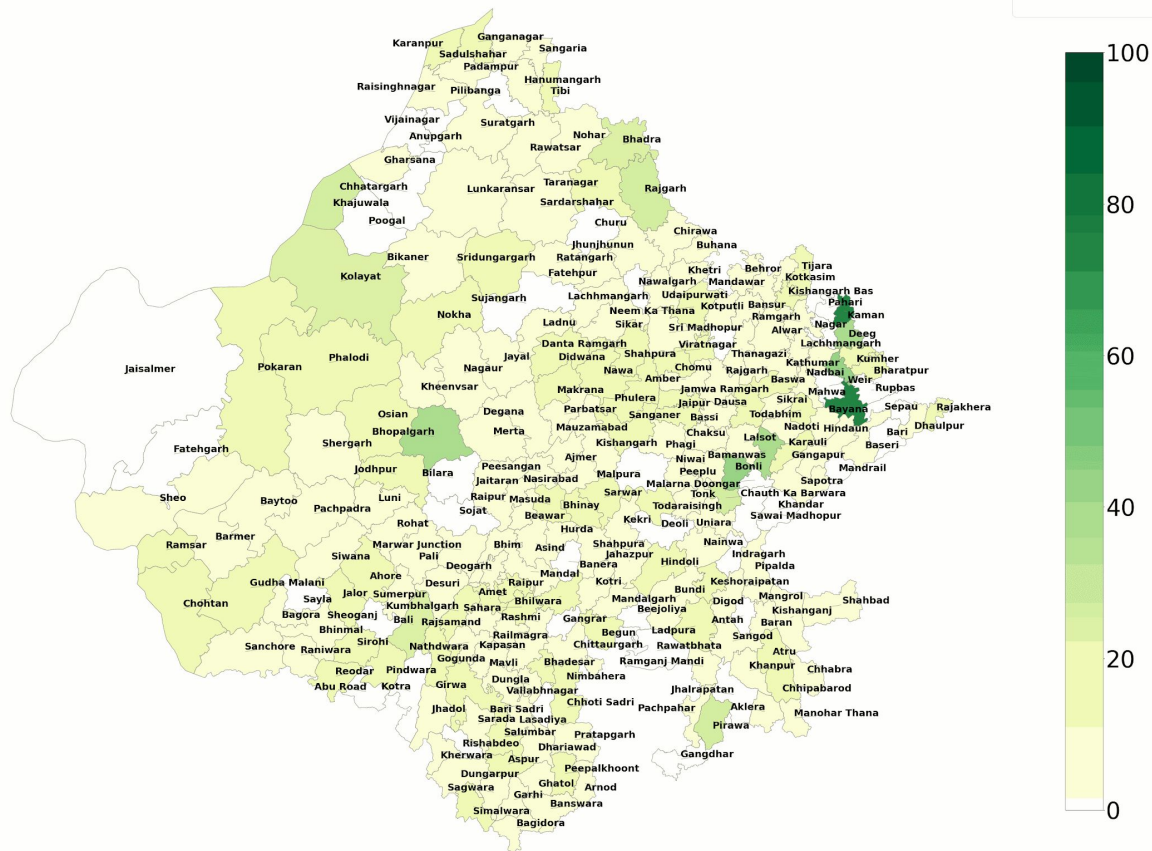


ASHA Segmentation



		Cluster Summary																																
Clusters	0	138	616	147	30	76	0	22	2	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	668	0	131	8	0	0	0
	1	13	55	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	24	0	13	0	0	0	0
	2	352	604	1665	768	572	24	217	22	52	8	0	0	0	2	3	0	3	0	887	6	345	472	6	20	3								
	3	231	082	672	137	201	6	87	6	18	2	0	0	0	2	2	0	2	0	1686	4	228	140	2	4	0								
	4	259	190	814	68	242	10	138	13	31	2	6	2	7	76	32	14	20	0	781	166	260	192	2	8	4								

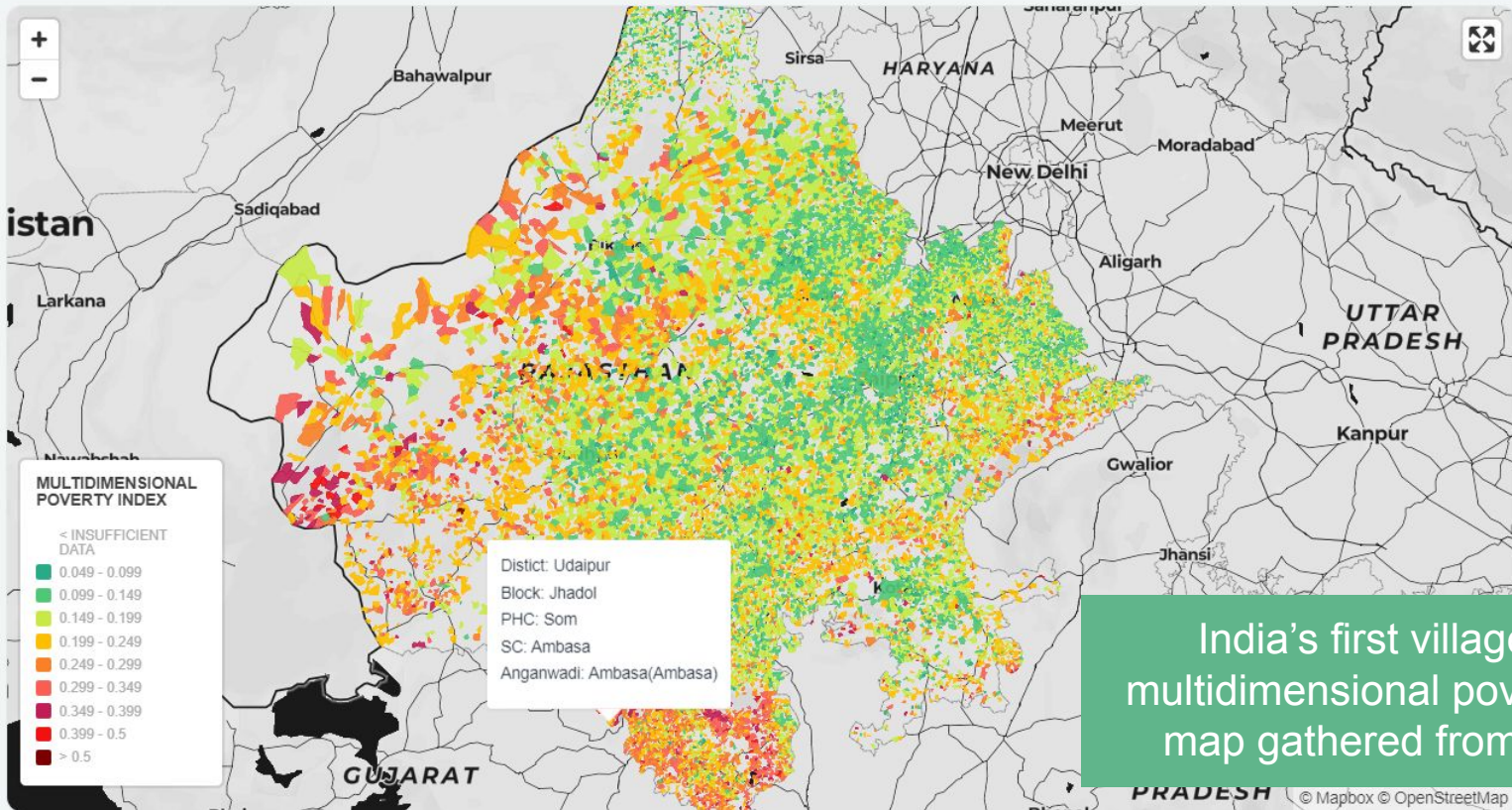
Khushi Baby INC, TSP, CHIP Block Level Analysis





Multidimensional Poverty Index

☐ Blank ☐ Light ☐ Dark



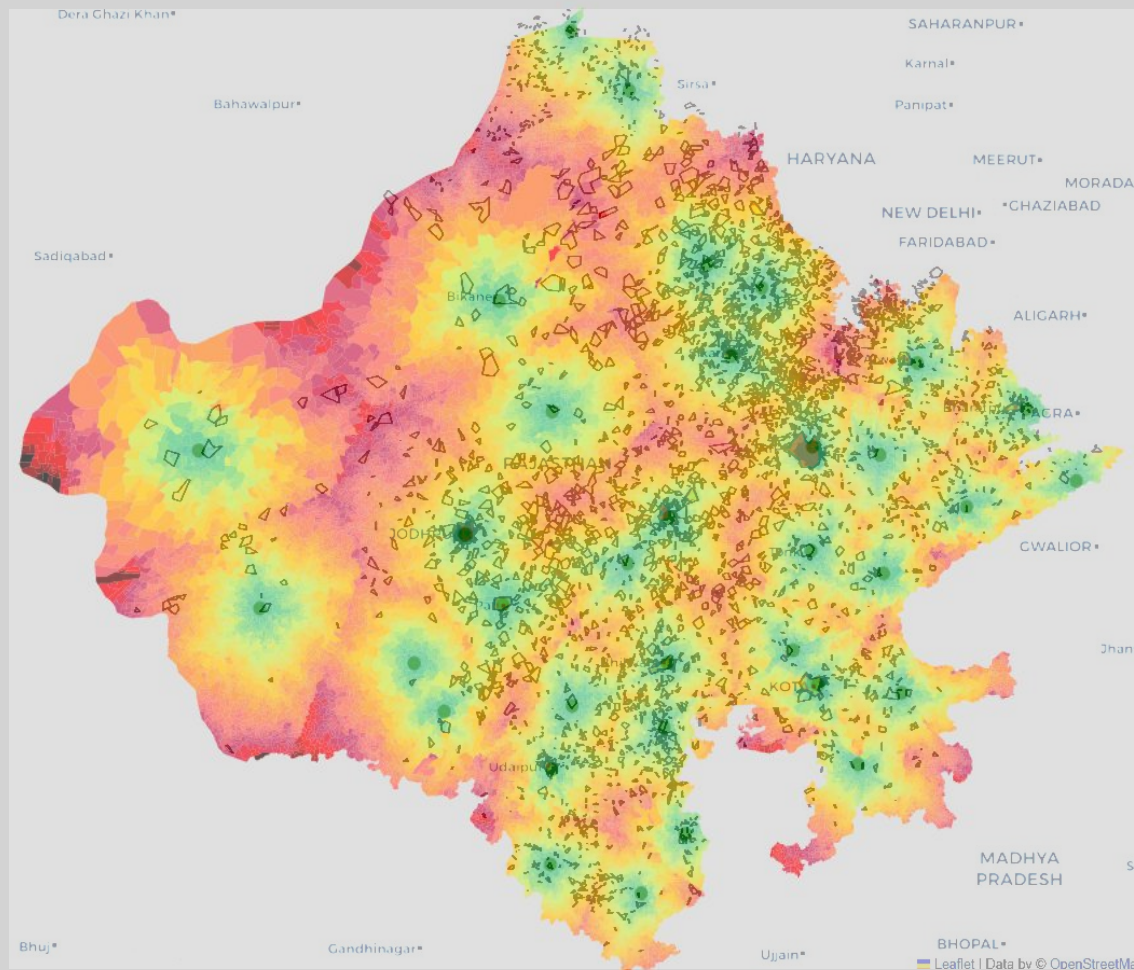
Distict: **Udaipur**
Block: **Jhadol**
PHC: **Som**
SC: **Kavel**
Anganwadi: **uplikavel(KAVEL)**
[Health Workers List](#)

Intensity score: **0.35%**
Multidimensional poverty indexing:
0.35%

Head count ratio (%): **1.00%**
Censored ☐ Uncensored ☐

- ☐ Nutrition: **0.00%**
- ☐ Years of Schooling: **6.00%**
- ☐ School Attendance: **6.00%**
- ☐ Cooking Fuel: **97.50%**
- ☐ Sanitation: **88.50%**
- ☐ Drinking Water: **74.00%**
- ☐ Electricity: **0.50%**
- ☐ Housing: **93.00%**
- ☐ Assets: **74.00%**

India's first village-level
multidimensional poverty index
map gathered from ASHAs



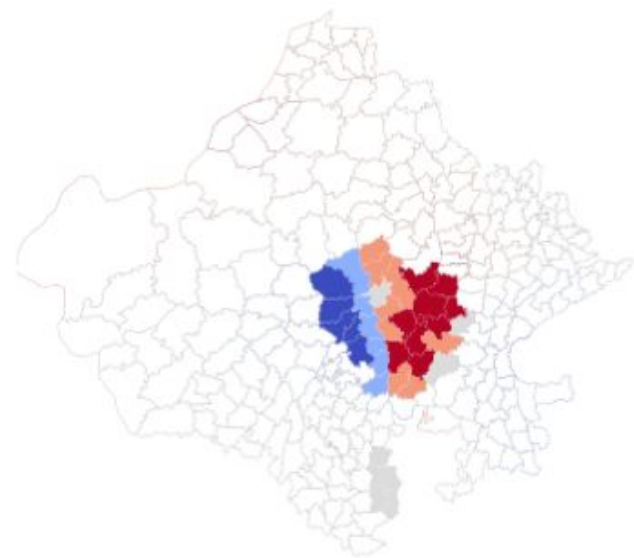
Accessibility vs health outcomes using geospatial insights to plan resource allocation

Driving time calculated from each of 42,000 villages to nearest medical college (red > 3 hours, green <30 minutes). Self-reported cardiovascular disease cases overlaid.

Useful for planning:

- > Cardiac care
- > Stroke care
- > Trauma
- > Emergency Obstetrics

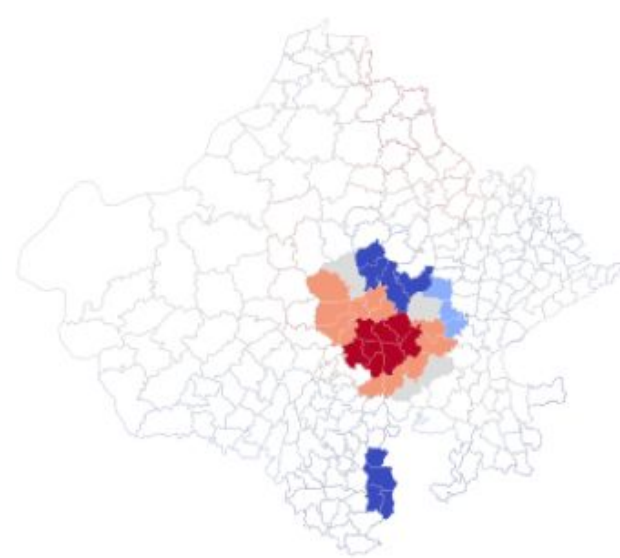
Local, spatial factors associated
with TB hotspots in Rajasthan to
fine-tune local campaigns



Malnutrition



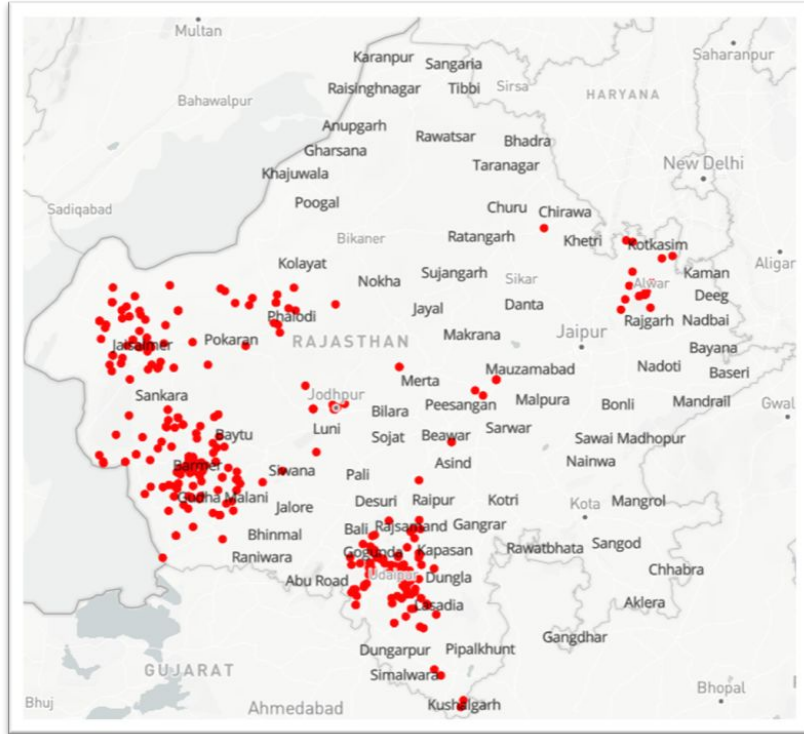
HIV



Brick Kiln

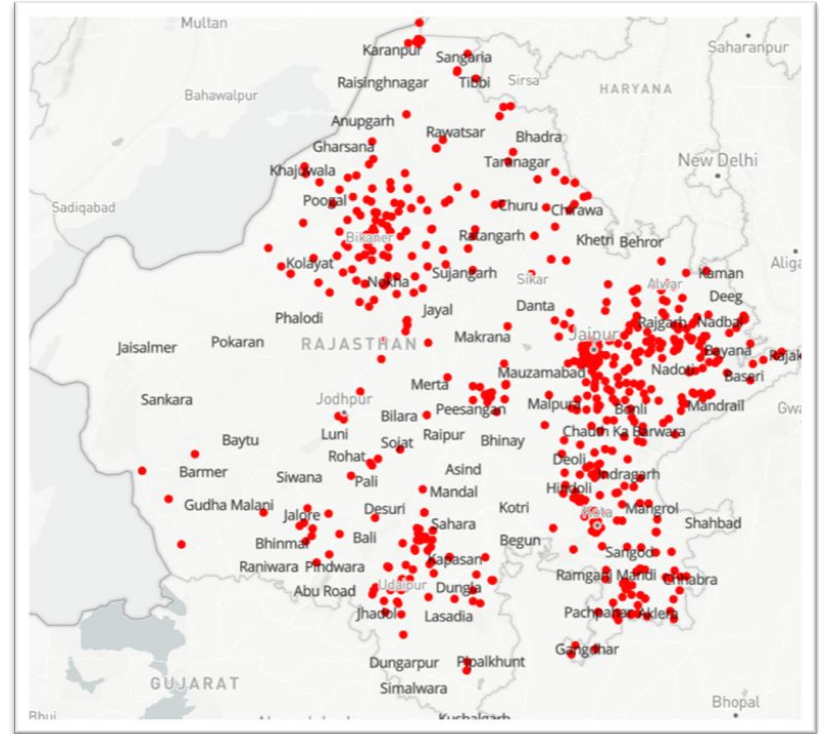
Monitoring of positive cases- Identification of Hotspots

Lab confirmed cases of Dengue and Malaria are being reverse geo-mapped in time series



Malaria

Vs

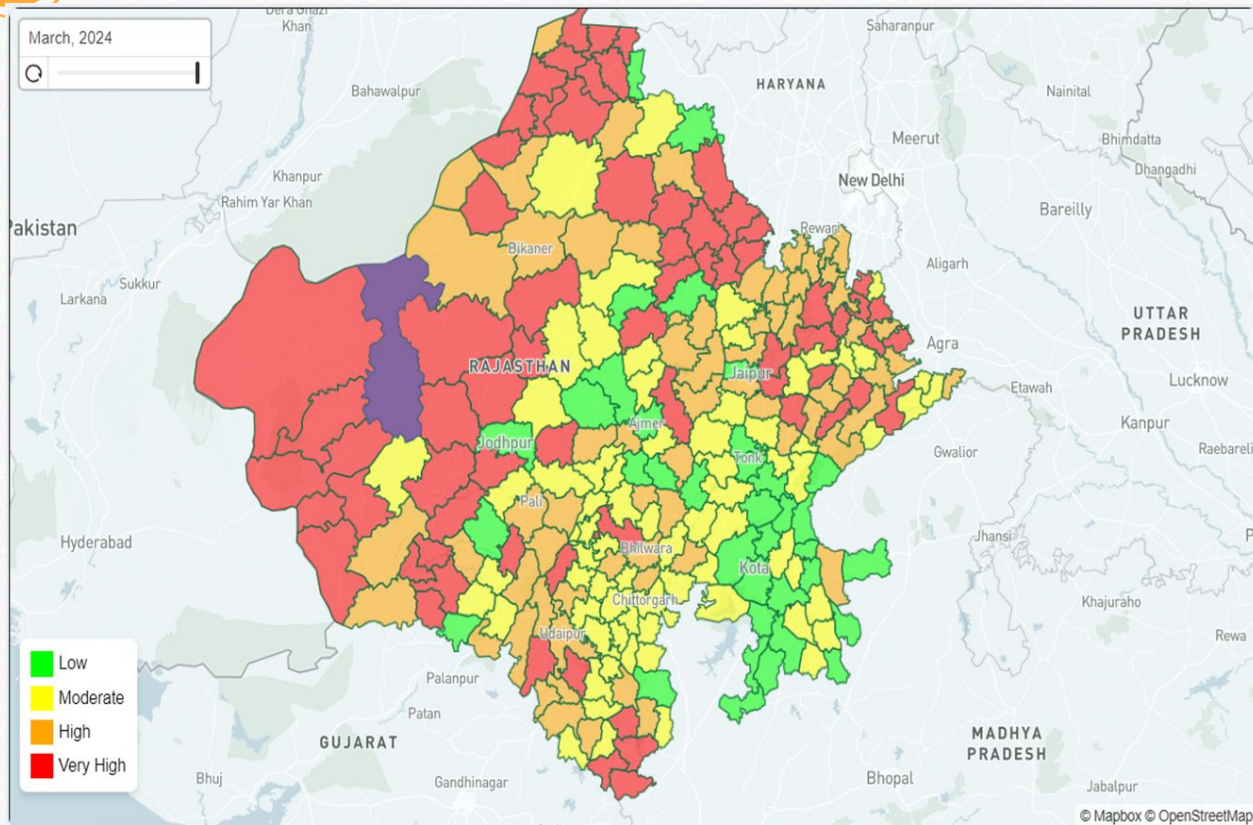


Dengue

SARIMA Model



CHVI Visualization up to Block level



District: Jaisalmer
Block: Pokaran
CHVI Score: 90%
Category: Very High

Contribution of Adaptive Capacity Variables: 7%

- Multidimensional Poverty Indexing (2%): 0.006 Dimensionless
- Caste ST (1%): 517 people
- No Formal Education (4%): 149 people
- Number of Beneficiaries Screened (23%): 17028 people
- Driving Time to Nearest Hospital (43%): 98 Minutes
- Normalized Data Quality Score (27%): 2.91

Contribution of Environmental Exposure Variables: 60%

- LST Minimum (6%): 26 °C
- LST Maximum (12%): 49 °C
- **LST Mean (11%): 39 °C**
- PM2.5 Minimum (2%): -1 µg/m³
- PM2.5 Maximum (12%): 2 µg/m³
- **PM2.5 Mean (7%): 0 µg/m³**
- CO Minimum (2%): 0.028 ppm
- CO Maximum (10%): 0.048 ppm
- **CO Mean (3%): 0.036 ppm**
- NO2 Minimum (6%): 0 ppb
- NO2 Maximum (7%): 0 ppb
- **NO2 Mean (2%): 0 ppb**
- SO2 Minimum (0%): -0.001 ppb
- SO2 Maximum (6%): 0.003 ppb
- **SO2 Mean (6%): 0 ppb**
- Rain Minimum (0%): 0 mm
- Rain Maximum (0%): 2 mm
- **Rain Mean (0%): 1 mm**

Maternal Anemia AI Model

Original
s7082247lv6gnbbeeh0_right_eye_3.png



Conjunctiva Region



Original
crkfptrlbu82cq82o0pg_right_eye_3.png



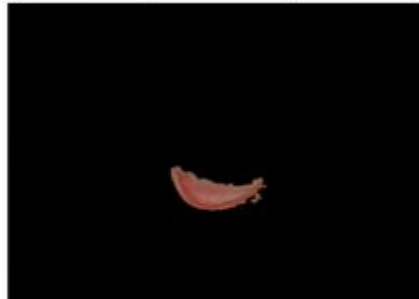
Conjunctiva Region



Original
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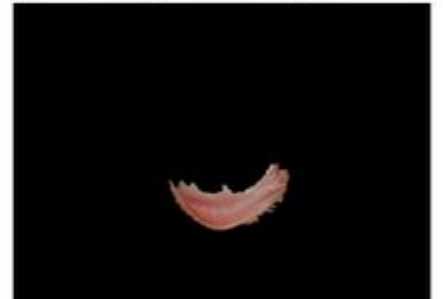
Conjunctiva Region



Original
cspele4vs8mnl4r5a9gg_right_eye_3.png



Conjunctiva Region



Q&A?

Thank You

Mohammad Sarfarazul Ambiya

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Climate Health Technical Advisor - JPAL

Great Learning Academic Mentor - MIT DSML

Data Science Mentor - Inttrvu.AI

Geospatial Scientist - OMDENA

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